



Assessment Project

Web Engineer Assessment Project v2.2.4

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1 Background

This document outlines the requirements for a mini-project designed to assess prospective Web Engineering candidates at 101 Digital.

1.1 Approach

Candidates undertaking this assessment are expected to carefully review and understand the following:

1. The project requirements detailed in this document.
2. The assessment criteria outlined below.
3. The deliverables to be submitted to 101 Digital.

1.2 Assessment Criteria & Deliverables

You will be assessed on the following assessment criteria:

1. Packaging and Delivery: The effectiveness of how the application, source code, and supporting documentation are packaged and delivered, ensuring a seamless review and testing experience.

2. Functional and Complete Solution: A fully functional application that meets the specified requirements. Candidates are welcome to implement additional features where they believe they provide meaningful value.

3. Architecture and Documentation: The quality and clarity of any documentation provided, including solution architecture, design decisions, assumptions, and any additional information that may assist reviewers in understanding the implementation.

4. Solution Demonstration: The ability to effectively present and demonstrate the solution to the 101 Digital Technical Evaluation Team, including key design decisions and implementation details.

5. Code Quality and Testing: Clean, maintainable, and well-structured code that adheres to industry best practices, supported by appropriate testing and quality assurance measures.

6. Value-Added Contributions: Additional enhancements, considerations, or innovations that extend beyond the stated requirements and demonstrate initiative, technical depth, or product thinking.

Send your submission by the due date and time, and any links to your code repositories to the following:

- a. ThanhNguyenBa@101digital.io
- b. rajiv@101digital.io.

2 Solution Requirements

You are required to design and develop a simple invoicing web application named ***SimpleInvoice***.

2.1 The *SimpleInvoice* Overview

The ***SimpleInvoice*** webapp application must provide the following core features:

1. User Authentication (Login):

- a. Implement the login functionality in accordance with the API specifications provided in **Appendix A**.
- b. Create a login screen/form that authenticates the user and securely stores the required information for subsequent API requests, including the access token and organisation token.
- c. Implement appropriate client-side validation for all login form fields.

2. Create Invoice:

- a. Implement invoice creation functionality in accordance with the API specifications provided in **Appendix A**.
- b. Create a form that allows users to enter invoice details.
- c. Each invoice should contain a single line item only.
- d. Implement validation for all relevant invoice fields.
- e. Submit the invoice data via the provided API endpoint. Upon successful submission, display a confirmation message notifying the user that the invoice has been created successfully.

3. List / Search / View Invoices:

- a. Create a screen that displays a list of invoices previously created through the application.
- b. Configure the invoice list page as the application's default landing (home) screen after authentication.
- c. Provide common data management capabilities, including:
 - Search
 - Sorting
 - Filtering
 - Pagination.

2.2 Technical Guidelines

The solution should adhere to the following technical requirements:

- Use **Next.js** and **TypeScript** for implementation.
- The application must be fully responsive and support a range of device sizes.
- You may use any styling framework or approach that you consider appropriate for delivering a professional user experience.
- The use of reputable open-source libraries is encouraged where they improve productivity, maintainability, or solution quality.
- Source code should be appropriately documented to facilitate understanding and review.

- Include a reasonable level of automated unit testing for key application functionality.

2.3 Security Considerations

As the solution is being developed using Next.js, which provides both client-side and server-side execution capabilities, candidates are expected to leverage the server-side runtime to implement secure handling of credentials, authentication tokens, and API communications.

Consideration should be given to industry best practices for authentication, authorization, token management, and the protection of sensitive information. Solutions that demonstrate a strong security posture and appropriate use of Next.js server-side features will be viewed favourably.

The following are strongly recommended and will be considered under Clean, quality, tested code and Value Add:

- **Server-side token exchange** — The `POST /oauth2/token` call (which requires `client_id`, `client_secret`, `username`, `password`) must be performed on the server (Route Handler / Server Action / API route), never from client-side code. The browser should call your own NextJS endpoint, which in turn calls the 101 Digital identity server.
- **Keep secrets out of the browser bundle** — Do not expose secrets via `NEXT_PUBLIC_*` environment variables (these are inlined into the client bundle). Secrets (`client_secret`, credentials) must be read only on the server from non-public environment variables.
- **Secure token storage** — Store the `access_token` and `org_token` in `httpOnly`, `Secure`, `SameSite` cookies set by the server, rather than `localStorage` / `sessionStorage` (which are readable by any script and vulnerable to XSS).
- **Backend-for-Frontend (BFF) / proxy pattern** — Route all calls to the membership-service and invoice-service through your NextJS server so that access tokens never reach the browser. The client should only ever talk to your own application's endpoints.
- **Secrets hygiene** — Keep secrets in a git-ignored `.env.local` and provide a committed `.env.example` with placeholder values. Do not commit real credentials to the repository.
- **Validation & headers** — Validate inputs on the server (not only via client-side form validation), and configure basic security headers (e.g. via `next.config.js` headers or middleware).

Important Note

- *The credentials provided in **Appendix A** are intended for use with a shared sandbox/development environment and are supplied solely to facilitate testing and evaluation.*
- *Although these credentials are non-production, they should be treated as sensitive information and managed in accordance with security best practices. Candidates are expected to load credentials and configuration values from environment variables or secure configuration mechanisms rather than hardcoding them within the application source code.*
- *Solutions that demonstrate secure handling of credentials and adherence to industry-standard security practices will be viewed favourably during the assessment process.*

3 Appendix A - APIs to be Used

3.1 Fetch App Access Token

POST https://is-wso2-dev.101digital.io/t/101digital.core/oauth2/token

Header

Content-Type: application/x-www-form-urlencoded

Content-type

application/x-www-form-urlencoded

Body

client_id= qIsGKsgR3Qt4M_oSAvRq2yChEpUa
 client_secret= GE7sxz9a4J6bw9LyPxkr4syV6pdLiMvYu2o_fDfnWgUa
 grant_type= password
 scope= openid
 username = 94756921275
 password= Password@12345

Parameters

client_id	qIsGKsgR3Qt4M_oSAvRq2yChEpUa
client_secret	GE7sxz9a4J6bw9LyPxkr4syV6pdLiMvYu2o_fDfnWgUa

Sample

```
curl --location 'https://is-wso2-dev.101digital.io/t/101digital.core/oauth2/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'client_id=qIsGKsgR3Qt4M_oSAvRq2yChEpUa' \
--data-urlencode 'client_secret=GE7sxz9a4J6bw9LyPxkr4syV6pdLiMvYu2o_fDfnWgUa' \
--data-urlencode 'grant_type=password' \
--data-urlencode 'scope=openid' \
--data-urlencode 'username=94756921275' \
--data-urlencode 'password=Password@12345'
```

A successful response will return an **access_token**, which you will need to use for the subsequent API calls.

3.2 Get user profile

GET https://api-neobank-dev.101digital.io/membership-service/1.0.0/users/me

Header

Authorization: Bearer **access_token**

Content-type

Content-Type: application/json

Sample

```
curl --location 'https://api-neobank-dev.101digital.io/membership-service/1.0.0/users/me' \
--header 'Authorization: Bearer
eyJ4NXQiOiJOVGRTWmpNNFpEazNOalkwWXpjNU1tWm1PRGd3TVRFRM01XWXdOREU1
TVdSbFpEZzROemM0WkEiLCJraWQiOiJNell4TW1Ga09HWXdNV0kwWldObU5EY3hORlI
3WW1NNFpUQTNNV0kyTkRBelpHUXpOR00wWkdSbE5qSmtPREZrWkRSaU9URmtNV0Z
oTXpVMlpHVmxOZ19SUzI1NiIsImFsZyI6IjJTMjU2In0.eyJzdWUiOiIzZTE2Y2VmZC01NjczLT
RjZGI0OGQ3Yy04YzJhZTg1MWVjNTMiLCJhdXQiOiJBUEFBMSUNBVEIPTI9VU0VSliwib3Jn
YW5pc2F0aW9uSWQiOiIxMGNhMTNmMS0xMTczLTQ2MzQtOTgxNy1iZDIzMmNINDgwMT
UiLCJpc3MiOiJodHRwczpcL1wwaXMtMTAxZGlnaXRhbC1zYW5kYm94LjEwMWRpZ2I0YW
wuaW86NDQzXC9vYXV0aDJcL3Rva2VuliwiZ3JvdXBzIjpbIkludGVybmFsXC9ldmVyeW9uZ
SIsIkludGVybmFsXC9zZWxmc2lnbnVwIl0slmdpdmluX25hbWUiOiJSdWtzaGFuliwiYXVkljoi
djNWODdaSXFqZGVNbIFsZjR5djdlVzNrMWFBYSlm5iZiI6MTcyOTc3Njk1MiwiYXpwIjoidjN
WODdaSXFqZGVNbIFsZjR5djdlVzNrMWFBYSlm5iZiI6MTcyOTc3Njk1MiwiYXpwIjoidjN
NTUyLCJpYXQiOiE3Mjk3NzY5NTIsImZhbWlseV9uYW1lIjoiUktTliwianRpljoiMmY2OTQ3Yz
gtOTM5NC00MDIwLTg3NGQtNmU2ZmE3NDEwOTNkIn0.KICy6vIGYhKw3M-
S8huhYBIW0fC4eGo0luuT9MqeTFju4uVnnsPJlt0dEUH5BPz7xSNfJWEYcVX4fWumTmVgl
hCJFN44FK51mSPNKFUZX57fg3D6ZLQnKj5WwMDe5B3Bl0Dpu8Z8NK3IBv3iO6H_ph_B
WXN4re1XGOoZbMMuUuFKVn0UtnuXjB1_IzGvKXnsuMEsTslbDcG0ewZwpK5cJrXHZ_zjs
Z-Llszpye6K0sEgwFyj0U1AEjKt0KgMFtDEKmkKuHoCnzopaNOK3TCa5Xe6zD8NMP-
lqtYq8uPVVsiiyLG_YHMOe0J98jKMNy1-uNMwJyNRAdiEnYu4uTLr1g
```

A successful response will return an **org_token (memberships[0].token)**, which you will need to use for the subsequent API calls.

3.3 Fetch Invoices API

GET

https://api-neobank-dev.101digital.io/invoice-
service/1.0.0/invoices?sortBy=CREATED_DATE&ordering=DESCENDING&pageNum=1&p
ageSize=10

Header

Authorization: Bearer **access_token**

org-token: **org_token**

Parameters

fromDate/toDate	represented in YYYY-MM-dd, i.e.: 2018-12-20
pageSize	number of records returned in a single request
pageNum	page number, starting with 1
ordering	Ordering result, i.e. ASCENDING or DESCENDING
sortBy	Sort the result, i.e. CREATED_DATE, etc
status	Filter the result, i.e. Paid
keyword	For search, i.e. 'IV1649318870503' <invoiceNumber>

Sample

```
curl --location 'https://api-neobank-dev.101digital.io/invoice-
service/1.0.0/invoices?sortBy=CREATED_DATE&ordering=DESCENDING&pageNu
m=1&pageSize=10' \
--header 'org-token:
eyJhbGciOiJIUzI1NiJ9.eyJvcmVudG90YXV1IjoiUnVrc2hhbiBTRCIsImZcyLl6ImIudGVybmF
sliwiaHR0cDpcL1wvd3NvMi5vcmdcL2NsYWltc1wvYXBwbGljYXRpb25uYW1lIjoiMT
AxRFBheUFwcClmIm1lbWJlcnNoaXBjZCI6ImU3OGU4M2JiLTViNjYtNDIyNi1hM2Jh
LWE1NTRhMTU1MzQ1ZCI6ImV4cCI6MTcyOTc4MDc0MSwidXNlcklkljoiNmRIMjM4
ZTgtZjc0Mi00NTQyLWUwZWVtYjIjNGEwNjIhZWVtZWVib3JnSWQiOiI0MGNhMTMh
MS0xMTczLTQ2MzQ0OTgxNy1iZDIzMmNINDgwMTUiLCJsaXN0Um9sZXMiOiI0MTJ3
nYW5pc2F0aW9uT3duZXIiXSwiaHR0cDpcL1wvd3NvMi5vcmdcL2NsYWltc1wvZW5
0aXR5SWQiOiI0MDFEIn0uodjAWDUHB5gsn48-N2ADfwwfDhZckLITLfEpTXcpl98' \
--header 'Authorization: Bearer
eyJ4NXQiOiJOVGRTWmpNNFpEazNOalkwWXpjNU1tWm1PRGd3TVRFRM01XWXd
OREU1TVdSbFpEZZROemM0WkEiLCJraWQiOiJNell4TW1Ga09HWXdNV0kwWldO
bU5EY3hOR1I3WW1NNFpUQTNNV0kyTkRBelpHUXpOR00wWkdSbE5qSmtPREZr
WkRSaU9URmtNV0Z0TXpVMlpHVmxvOZ19SUzI1NiIsImFsZyI6IlJTMjU2In0ueyJzdW
liOiIzZTE2Y2VmZC01NjczLTRjZGltOGQ3Yy04YzJhZTg1MWVjNTMiLCJhdXQiOiJB
UFBMSUNBVEIPTI9VU0VSliwib3JnYW5pc2F0aW9uSWQiOiI0MGNhMTMhMS0xMT
TczLTQ2MzQ0OTgxNy1iZDIzMmNINDgwMTUiLCJpc3MiOiI0JodHRwczpcL1wvaXMtM
TAXZGlnaXRhbC1zYW5kYm94LjEwMWRpZ2I0YWwuaW86NDQzXC9vYXV0aDJsL
3Rva2VuliwiZ3JvdXBzIjpbIkludGVybmFsXC9ldmVyeW9uZSIsIkludGVybmFsXC9zZ
Wxmc2lnbnVwll0sImdpdmVuX25hbWUiOiJSdWtzaGFuIiwiaXVkljoidjNWODdaSXFq
```


ZFVNblFsZjR5djdIVzNrMWFBYSIsIm5iZil6MTcyOTc3Njk1MiwiYXpwljoidjNWODda
SXFqZFVNblFsZjR5djdIVzNrMWFBYSIsInNjb3BlIjoib3BlbmklwiZXhwIjoxNzI5Nzgw
NTUyLCJpYXQiOjE3Mjk3NzY5NTIsImZhbWlseV9uYW1lIjoUktTliwianRpljoiMmY2O
TQ3YzgtOTM5NC00MDIwLTg3NGQtNmU2ZmE3NDExOTNkIn0.KICy6vIGYhkW3M

-
S8huhYBIW0fC4eGo0luuT9MqeTFju4uVnnsPJlt0dEUH5BPz7xSNfJWEYcVX4fWu
mTmVglhCJFN44FK51mSPNKFUZX57fg3D6ZLQnKj5WwMDe5B3BloDpu8Z8NK3I
Bv3iO6H_ph_BWXN4re1XGOoZbMMuUuFKVn0UtnuXjB1_lzGvKXnsuMEsTslbDcG
0ewZwpK5cJrXHZ_zjsZ-

Llszpye6K0sEgwFyj0U1AEjKt0KgMFtDEKMKuHoCnzopaNOK3TCa5Xe6zD8NMP-
lqtqYq8uPvVsiyiLG_YHMOe0J98jKMny1-uNMwJyNRAdiEnYu4uTLr1g' \

--header 'Cookie: JSESSIONID=CD19BD436FA70D1975EDC6FF8A873E1A'

3.4 Create Invoice API

POST

https://api-neobank-dev.101digital.io/invoice-service/1.0.0/invoices

Header

Authorization: Bearer **access_token**

Operation-Mode: **SYNC**

org-token: **org_token**

Content-type

application/json

Body

```
{
  "invoices": [
    {
      "bankAccount": {
        "bankId": "",
        "sortCode": "09-01-01",
        "accountNumber": "12345678",
        "accountName": "John Terry"
      },
      "customer": {
        "firstName": "Nguyen",
        "lastName": "Dung 2",
        "contact": {
          "email": "nguyendung2@101digital.io",
          "mobileNumber": "+6597594971"
        }
      },
      "addresses": [
        {
          "premise": "CT11",
          "countryCode": "VN",
          "postcode": "1000",
          "county": "hoangmai",
          "city": "hanoi",
          "addressType": "BILLING"
        }
      ]
    }
  ],
  "documents": [
    {
      "documentId": "96ea7d60-89ed-4c3b-811c-d2c61f5feab2",
      "documentName": "Bill",
      "documentUrl": "http://url.com/#123"
    }
  ]
}
```

```

    }
  ],
  "invoiceReference": "#123456",
  "invoiceNumber": "INV123456701",
  "currency": "GBP",
  "invoiceDate": "2021-05-27",
  "dueDate": "2021-06-04",
  "description": "Invoice is issued to Akila Jayasinghe",
  "customFields": [
    {
      "key": "invoiceCustomField",
      "value": "value"
    }
  ],
  "extensions": [
    {
      "addDeduct": "ADD",
      "value": 10,
      "type": "PERCENTAGE",
      "name": "tax"
    },
    {
      "addDeduct": "DEDUCT",
      "type": "FIXED_VALUE",
      "value": 10.00,
      "name": "discount"
    }
  ],
  "items": [
    {
      "itemReference": "itemRef",
      "description": "Honda RC150",
      "quantity": 1,
      "rate": 1000,
      "itemName": "Honda Motor",
      "itemUOM": "KG",
      "customFields": [
        {
          "key": "taxiationAndDiscounts_Name",
          "value": "VAT"
        }
      ]
    },
    {
      "addDeduct": "ADD",
      "value": 10,
      "type": "FIXED_VALUE",
      "name": "tax"
    }
  ]

```

```

    },
    {
      "addDeduct": "DEDUCT",
      "value": 10,
      "type": "PERCENTAGE",
      "name": "tax"
    }
  ]
}
]
}
]
}
}

```

Sample

```

curl --location 'https://api-neobank-dev.101digital.io/invoice-service/1.0.0/invoices' \
--header 'org-token:
eyJhbGciOiJIUzI1NiJ9.eyJvcmdOYW1lIjo1UnVrc2hhbiBTRCIslmIzcyI6ImIudGVybmFsliwiaHR0cDpcL1wvd3NvMi5vcmdcL2NsYWltc1wvYXBwbGljYXRpb25uYW1lIjo1MTAxRFBheUFwcClslm1lbWJlcnNoaXBjZCI6ImU3OGU4M2JiLVNiYtNDIyNi1hM2JhLWE1NTRhMTU1MzQ1ZCIslmV4cCI6MTcyOTc0MDc0MSwidXNlckkljoiNmRlMjM4ZTgtZjc0Mi00NTQyLWEwZWMTYjIjNGEwNjIhZWwliwib3JnSWQiOiIxMGNhMThmMS0xMTczLTQ2MzQtOTgxNy1iZDIzMmNINDgwMTUiLCJsaXN0Um9sZXMiOiSiT3JnYW5pc2F0aW9uT3duZXliXSwiaHR0cDpcL1wvd3NvMi5vcmdcL2NsYWltc1wvZW50aXR5SWQiOiIxMDFEIn0.odjAWDUHB5gsn48-N2ADfwwfDhZckLITLfEpTXcpl98' \
--header 'Operation-Mode: SYNC' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
eyJ4NXQiOiJOVGRTWmpNNFpEazNOalkwWXpjNU1tWm1PRGd3TVRFRM01XWXdOREU1TVdSbFpEZzROemM0WkEiLCJraWQiOiJNell4TW1Ga09HWXdNV0kwWldObU5EY3hORlI3WW1NNFpUQTNNV0kyTkRBelpHUXpOR00wWkdSbE5qSmtPREZrWkRSaU9URmtNV0ZoTXpVMlplHVmxOZ19SUzI1NiIsImFsZyI6IiJTMjU2In0.eyJzdWliOiIzZTEyY2VmZC01NjcZLTJjZGltOGQ3Yy04YzJhZTg1MWVjNTMiLCJhdXQiOiJBUFBMSUNBVEIPTl9VU0VSliwib3JnYW5pc2F0aW9uSWQiOiIxMGNhMThmMS0xMTczLTQ2MzQtOTgxNy1iZDIzMmNINDgwMTUiLCJpc3MiOiJodHRwczpcL1wvaXMtMTAxZGlnaXRhbC1zYW5kYm94LjEwMWRpZ2I0YWwuaW86NDQzXC9vYXV0aDJsL3Rva2VuliwiZ3JvdXBzIjpbIkludGVybmFsXC9ldmVyeW9uZSIsIkludGVybmFsXC9zZWxmc2lnbnVlI0slmdpdmVuX25hbWUiOiJScDwtzaGFuIiwiaXVkljoiZjNWODdaSXFqZFVNbIFsZjR5djdlVzNrMWFBYSIsIm5iZiI6MTcyOTc3Njk1MiwiYXpwIjojdjNWODdaSXFqZFVNbIFsZjR5djdlVzNrMWFBYSIsInNjb3BlbmkiIiwiaXhwIjojZl5NzgwNTUyLCJpYXQiOiE3Mjk3NzY5NTIsImZhbmVseV9uYW1lIjo1UktTliwianRpljoiMmY2OTQ3YzgtOTM5NC00MDIwLTg3NGQtNmU2ZmE3NDExOTNkIn0.KICy6vIGYhkW3M-S8huhYBIW0fC4eGo0luuT9MqeTFju4uVnnsPJlt0dEUH5BPz7xSNfJWEYcVX4fWumTmVglhCJFN44FK51mSPNKFUZX57fg3D6ZLQnKj5WwMDe5B3BlOdpU8Z8NK3IBv3iO6H_ph_BWXN4re1XGOoZbMMuUuFKVn0UtnuXjB1_lzGvKXnsuMEsTslbDcG0ewZwpK5cJrXHZ_zjsZ-Llszpye6K0sEgwFyj0U1AEjKt0KgMfTDEKMKuHoCnzopaNOK3TCa5Xe6zD8NMP-lqtYq8uPVVsiyiLG_YHMOe0J98jKMNy1-uNMwJyNRAdiEnYu4uTLr1g' \
--data-raw '{
  "invoices": [

```

```

{
  "bankAccount": {
    "bankId": "",
    "sortCode": "09-01-01",
    "accountNumber": "12345678",
    "accountName": "John Terry"
  },
  "customer": {
    "firstName": "Nguyen",
    "lastName": "Dung 2",
    "contact": {
      "email": "nguyendung2@101digital.io",
      "mobileNumber": "+6597594971"
    },
    "addresses": [
      {
        "premise": "CT11",
        "countryCode": "VN",
        "postcode": "1000",
        "county": "hoangmai",
        "city": "hanoi",
        "addressType": "BILLING"
      }
    ]
  },
  "documents": [
    {
      "documentId": "96ea7d60-89ed-4c3b-811c-d2c61f5feab2",
      "documentName": "Bill",
      "documentUrl": "http://url.com/#123"
    }
  ],
  "invoiceReference": "#123456",
  "invoiceNumber": "INV123456701",
  "currency": "GBP",
  "invoiceDate": "2021-05-27",
  "dueDate": "2021-06-04",
  "description": "Invoice is issued to Akila Jayasinghe",
  "customFields": [
    {
      "key": "invoiceCustomField",
      "value": "value"
    }
  ],
  "extensions": [
    {
      "addDeduct": "ADD",
      "value": 10,

```

```

        "type": "PERCENTAGE",
        "name": "tax"
    },
    {
        "addDeduct": "DEDUCT",
        "type": "FIXED_VALUE",
        "value": 10.00,
        "name": "discount"
    }
],
"items": [
    {
        "itemReference": "itemRef",
        "description": "Honda RC150",
        "quantity": 1,
        "rate": 1000,
        "itemName": "Honda Motor",
        "itemUOM": "KG",
        "customFields": [
            {
                "key": "taxiationAndDiscounts_Name",
                "value": "VAT"
            }
        ],
        "extensions": [
            {
                "addDeduct": "ADD",
                "value": 10,
                "type": "FIXED_VALUE",
                "name": "tax"
            },
            {
                "addDeduct": "DEDUCT",
                "value": 10,
                "type": "PERCENTAGE",
                "name": "tax"
            }
        ]
    }
]
}

```